יום ניצוץ!
הפרויקטים הכי מבריקים בפקולטה

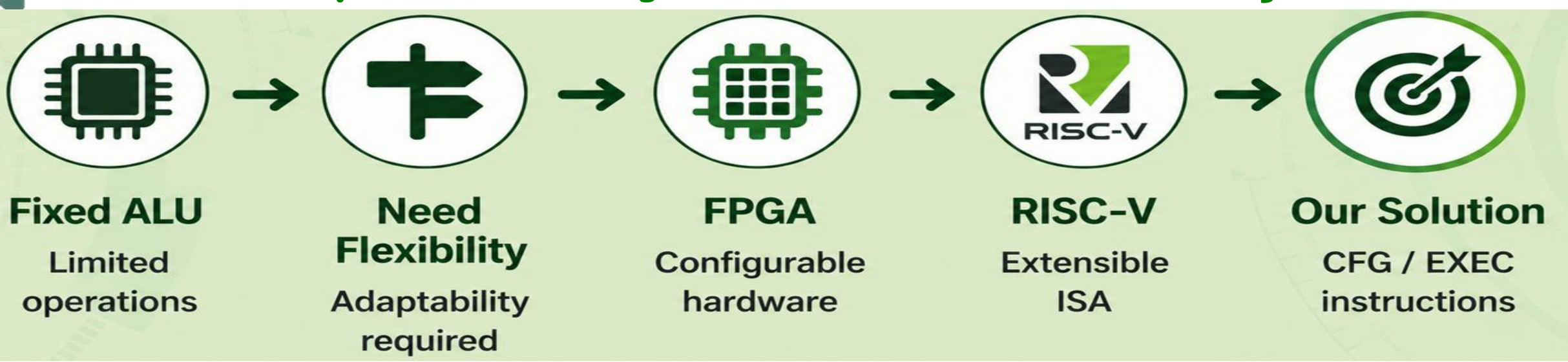
הפרויקטים הכי מבריקים בפקולטה

VLSI

הפקולטה להנדסת חשמל ומחשבים
ע"ש אנדרו וארנה ויטרבי

Development and Implementation of an FPGA Block on a RISC-V Processor

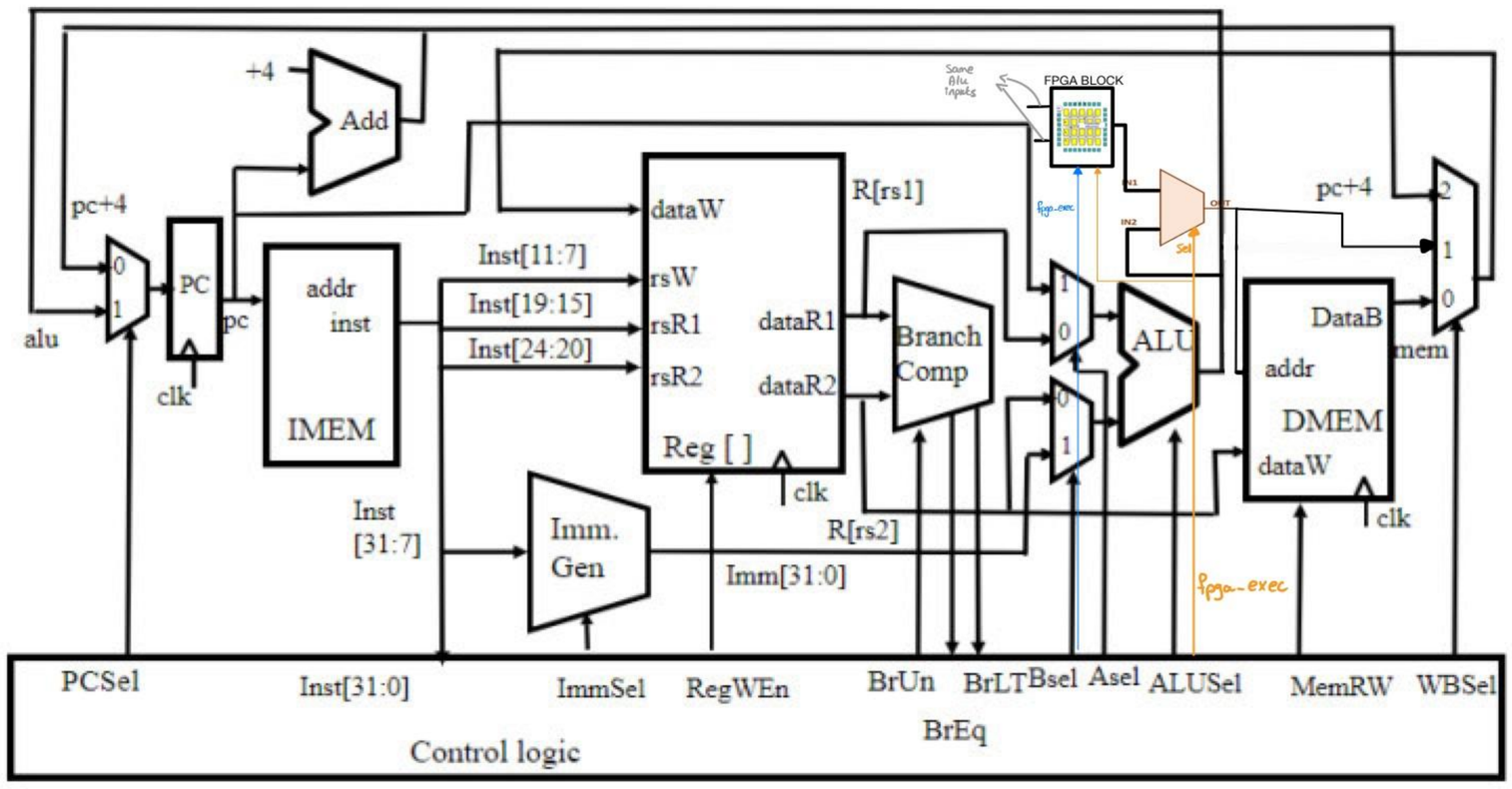
Hala Rabah & Deema khalaily
Supervised by Amnon Stanislavsky



Project description

Designed and integrated a configurable FPGA fabric into a custom single-cycle RISC-V processor. Implemented custom CFG and EXEC instructions for runtime FPGA configuration and execution using SystemVerilog. Verified functionality with modular testbenches, **VCS** simulations, and **Verdi** waveform analysis. Designed the FPGA architecture in draw.io and implemented RTL modules including LUTs, routing, and configuration control. Performed synthesis with **Synopsys Design Vision** and physical design exploration with **Cadence Innovus**. Demonstrated successful integration of FPGA-accelerated operations alongside standard RISC-V ALU execution.

PROCESSOR'S ARCHITECTURE



CFG & EXEC Operations

1 CFG Instruction
Opcode 0x7F, funct3=0. Two CFGs to complete the 62-bit configuration.

2 Configuration Controller
Captures and stores configuration data. cfg_done signals completion before execution.

3 EXEC Instruction
Opcode 0x7F, funct3=1. Operands are read from the register file.

4 Write-Back Integration
MUX selects ALU or FPGA result. Result written back to register.

Synthesis & Physical Design

RTL to Gates

Synopsys Design Vision Synthesized SV RTL → Gate-level netlist

Physical Design

Cadence Innovus Place & Route flow: floorplan, CTS, routing

FPGA ARCHITECTURE

